SE4AI - Requirements Level Explanations for Different Stakeholders

Hannah Estes
Mohsen EsfandyariDoulabi
Saba Khanmohammadi

ACM Reference Format:
Hannah Estes, Mohsen EsfandyariDoulabi, and Saba Khanmohammadi. 2026. SE4AI - Requirements Level Explanations for Different Stakeholders. In . ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

Software engineers routinely face decisions that require balancing competing objectives, such as faster delivery versus lower cost, security versus usability, and accuracy versus interpretability. Multi-objective optimization offers principled methods for navigating these trade offs, and repositories such as MOOT [40] now provide over 120 real world SE optimization tasks spanning software configuration, cloud tuning, project health, and process modeling. The algorithmic machinery exists; the interpretation problem does not. When an optimizer produces a Pareto front or a decision tree over configuration parameters, a developer or project manager sees numeric trade offs between objective values, throughput versus latency, effort versus defect risk. What they do not see is what those trade-offs mean at the level of requirements: which stakeholder concerns are being satisfied, which are being compromised, and why a particular configuration represents a defensible decision. This interpretation gap between optimization outputs and requirement-level reasoning is a persistent barrier to adoption. As Menzies et al. observe, industrial practitioners continue to deliver sub-optimal products in part because they lack the tools to explore and understand trade offs.

Recent work in explainable AI (XAI) has made progress on translating model outputs into natural language, and large language models (LLMs) have shown promise as verbalization engines for decision-tree explanations [52]. Separately, the requirements engineering literature has long recognized that stakeholders reason about trade-offs differently depending on their role, priorities, and technical background [27]. However, no prior work combines these threads: generating stakeholder-tailored, requirement-level explanations of multi-objective optimization outputs, and evaluating whether such explanations actually improve trade off reasoning. This paper addresses that gap. We present a framework that maps multi-objective optimization outputs from MOOT tasks to requirement-level explanations via a structured schema and LLM-based verbalization. We evaluate explanation quality along three dimensions: faithfulness, clarity, and consistency, and assess whether tailoring explanations to distinct stakeholder profiles changes how trade offs are understood and acted upon. Specifically, we ask:

- **RQ1:** How can multi-objective optimization outputs be systematically translated into requirement-level explanations?
- **RQ2:** Do requirement-level explanations improve alignment between stakeholder priorities and optimization decisions?
- **RQ3:** What failure modes emerge when LLMs mediate optimization explanations?

To the best of our knowledge, this is the first work to combine LLM-based explanation generation, multi-objective optimization output, multi-stakeholder tailoring, and evaluation within a software engineering context.

2 A Gap in the Literature

Once establishing the motivation of our study, we investigated the current literature at the intersection of LLM-based explanation generation, multi-objective optimization, requirements engineering, and human-centered decision making. To do so, we followed a similar method outlined in a similar study [51]. This literature selection procedure follows four steps: 1) identify search queries for Google Scholar in the past decade, 2) rank output by citation counts, 3) calculated the *knee* of the citation curve (e.g., Figure 1), which is a frequently used literature review heuristic [10, 57], where papers above this point represent more groundbreaking and seminal work [45]. As noted by Senthilkumar and Menzies, this process allows the research team to prioritize the most influential papers reaching a state of *conceptual saturation* [22].

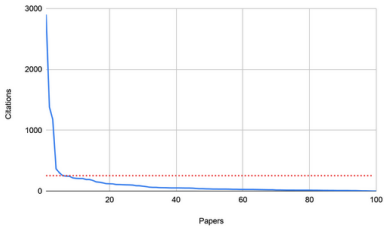

Figure 1: Citation curve for the top 100 papers returned by the query *human-AI decision making from 2015* on scholar.google.com (retrieved in 2026). The vertical line indicates the *knee* of the curve (i.e., 253 citations) defined as the furthest point to a line connecting the first and last points.

For this process, an initial eight queries were selected resulting in 734 papers, where a subset of 87 papers met the knee threshold. The

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
Conference'17, Washington, DC, USA
© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

acceptable papers' abstracts and titles were then read to confirm their relevance to our research questions and goals. Additional papers were found using snowballing [28] to fill additional gaps in the current review. This refinement of papers resulted in a final set of 35 papers. For the majority of these papers, the research team identified some common criteria that directly informed our research questions:

(1) **LLM corpus generation?** There a small set of papers that use LLMs to simulate personas which provides direct validation for our precursor step. Similarly, prior work has used LLMs to simulate stakeholder voices in requirements engineering specific tasks. No prior work generated structured persona-annotated output sets as a precursor to multi-objective explanation generation.
(2) **LLM explanation optimization?** This cluster of papers identifies prior work that is most similar to ours. By using explainable-AI to generate natural language explanations of decision-tree outputs or generating audience-tailored summaries, there is no emphasis on optimizing these prompts based on stakeholders from a requirements or software engineering specific context.
(3) **Optimization or decision-tree output explained?** This cluster of papers targets decision-tree outputs, but none are related to software or requirements engineering, differentiate by a stakeholder's background, and only one conducts a formal human acceptance study. The only prior work which combines multi-objective optimization with stakeholder decision support was conducted prior to the large popularity of LLMs.
(4) **Multi-stakeholder tailoring?** The need for stakeholder-specific explanations is becoming more popular in the explainable-AI (XAI) research field, but is largely underexplored.
(5) **Human acceptance study?** Studies related to a human's trust and acceptance of LLMs and AI are well studied, however there is no prior work on how different stakeholders accept or trust LLM generated information about optimization tradeoffs.
(6) **Requirements Engineering & Software Engineering context?** The requirements engineering-specific literature primarily handles stakeholder trade-offs conceptually, however the papers noted here identify the rapidly growing research area of large language models for requirements engineering (LLM4RE). There has been no prior work on LLM generated explanations for different stakeholders within a requirements and software engineering context.

Table 1 clusters papers (i.e., citation reference, year, and citation count) by their respective query and their relevance to the research question; for brevity, papers are not included if the do not directly relate to a research question. This table and the assoiated venn diagram identify the clear gap that this paper aims to fill. The prior work does not combine the six criteria we have identified simultaneously.

3 Background

3.1 Motivation: Validity of the Problem

The interpretation gap this paper addresses is grounded in a well-documented pattern in human-AI interaction: providing a decision does not equate to supporting the reasoning behind it. Amershi et al. [2] identify *making clear what the system can and cannot do* and *explaining why a recommendation was made* as two of eighteen foundational guidelines for human-AI interaction, noting that violations of these guidelines consistently degrade trust and adoption. Yet optimization outputs in software engineering—Pareto frontiers, decision trees, win-score tables—routinely present results without either.

The consequences of this gap are concrete. Research on human-AI decision-making shows that when outputs lack interpretable justification, users either over-rely on the AI regardless of its accuracy, or reject it entirely due to lack of trust [5, 16]. Buçinca et al. [5] demonstrate that cognitive forcing functions—interventions that require users to reason before accepting a recommendation—improve decision quality by reducing automation bias, but only when the underlying explanation is comprehensible. Without comprehension, forced reflection adds friction without benefit. More recently, Lee et al. [34] report that generative AI tools are already reducing the critical thinking effort practitioners invest in AI-assisted decisions, with self-reported confidence declining when outputs are opaque. These findings converge on a single implication: the quality and tailoring of explanations matters as much as the quality of the underlying recommendations.

Trust calibration provides a second motivation. A large meta-analysis by Kaplan et al. [29] ($k = 74$ studies) finds that *transparency*—the degree to which an AI's reasoning is visible and interpretable—is among the strongest predictors of appropriate trust in AI systems, outperforming performance accuracy alone in several domains. Rheu et al. [47] extend this to conversational and explanatory agents, finding that stakeholders develop trust through accumulated interactions with clear, role-relevant explanations rather than through a single high-accuracy output. In a multi-objective optimization context, where no single "correct" answer exists and every choice involves trade-offs, trust calibration is especially critical: a Project Manager who blindly accepts the highest-accuracy tree and a Software Engineer who rejects the simplest tree because they cannot interpret its stability score both represent calibration failures.

The XAI literature further validates the specific form of our intervention. Shin [54] distinguishes *explainability* (the degree to which a model's outputs can be described) from *causability* (the degree to which those descriptions enable human causal reasoning), finding that causability independently predicts acceptance beyond explainability alone. Dwivedi et al. [20] and Wang et al. [58] both argue that XAI systems must be designed around specific user mental models rather than generic natural-language translations. Kim et al. [31] provide direct empirical support for stakeholder-tailored XAI interfaces, showing that role-specific explanations improve perceived usefulness and decision confidence compared to one-size-fits-all presentations. Dhar et al. [17] make the normative case even more sharply, arguing that most XAI research remains "technocratic"—optimizing for technical fidelity without accounting

Citation	Year	Citations	1. LLM Persona	2. XAI Opt.	3. Opt. DTs	4. Stakeholder	5. Human Study	6. RE & SE
[2]	2019	2903	✓	-	-	-	✓	-
[55]	2019	1382	-	-	-	-	✓	-
[36]	2019	1293	-	-	-	-	-	-
[5]	2021	1177	-	-	-	-	✓	-
[16]	2016	869	-	-	-	-	✓	-
[34]	2025	515	✓	-	-	-	-	-
[18]	2019	498	-	✓	-	-	✓	-
[29]	2023	460	-	-	-	-	✓	-
[6]	2020	453	-	✓	-	-	✓	-
[47]	2021	414	-	-	-	-	✓	-
[9]	2018	823	✓	✓	-	-	-	-
[8]	2023	178	-	✓	-	-	-	-
[54]	2021	1603	-	✓	-	-	✓	-
[20]	2023	1513	-	✓	-	-	-	-
[58]	2019	1265	-	✓	-	-	✓	-
[59]	2019	1197	-	✓	-	-	-	-
[42]	2021	1083	-	✓	-	-	✓	-
[19]	2017	816	-	✓	-	-	-	-
[23]	2018	509	-	✓	-	-	-	-
[21]	2005	260	-	-	-	-	-	✓
[14]	2005	186	-	-	-	-	-	✓
[15]	2003	163	-	-	-	-	-	✓
[53]	2014	154	-	-	✓	-	-	✓
[1]	2002	141	-	-	-	-	-	✓
[27]	2018	159	-	-	-	-	-	✓
[26]	2018	137	-	-	-	-	-	✓
[48]	2015	77	-	-	-	-	-	✓
[49]	2017	56	-	-	-	✓	-	✓
[43]	2017	54	-	-	-	✓	-	✓
[25]	2023	315	-	✓	-	-	✓	-
[38]*	2024	93	✓	✓	-	-	-	-
[31]*	2024	68	-	-	-	✓	-	-
[24]*	2023	317	-	-	-	✓	✓	-
[41]*	2024	16	-	-	-	✓	-	-
[17]*	2025	1	-	-	-	✓	-	-
[52]*	2024	7	✓	✓	✓	-	-	-
[62]*	2023	28	-	✓	✓	-	-	-
[56]*	2023	16	-	✓	✓	-	✓	-
[50]*	2024	86	✓	-	-	✓	✓	-
[3]*	2024	243	✓	-	-	-	-	✓
[60]*	2025	11	✓	-	-	-	-	✓
[46]*	2025	37	-	✓	-	-	-	-
		Our Paper	✓	✓	✓	✓	✓	✓

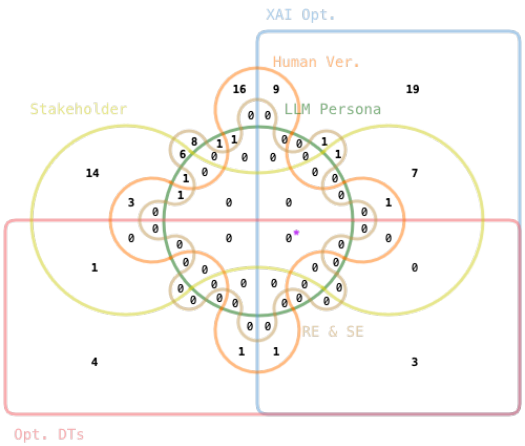

Table 1: Literature Review Results. An * denotes a new paper, and a - denotes no information related to that category in this paper. Each column in the table correlates to a category (number matches description in Section 2). The right hand side is a Venn diagram representation showing the gap our paper aims to fill denoted by a purple star.

for who the explanation is for, what they need to do with it, and how it fits into their existing reasoning processes.

Taken together, this body of work establishes that the problem is real (opaque optimization outputs damage trust and decision quality), that the solution space is tractable (role-tailored natural-language explanations improve calibration), and that no prior work has applied these principles to the specific challenge of multi-objective optimization outputs in software engineering contexts.

3.2 Origin of Data and Participants

Optimization task and data. Our study uses the MOOT repository (Multi-Objective Optimization Tasks) [40], a publicly available benchmark of over 120 real-world software engineering optimization tasks spanning software configuration, cloud resource tuning, project health prediction, and process modeling. We select the *project health* task, which uses the A2C reinforcement learning algorithm applied to software project health data (223 configurations, 10 features, 3 objectives: maximizing evaluation mean return, minimizing evaluation variance, and minimizing convergence time). This task was selected because its three-objective Pareto frontier produces interpretable trade-offs that are relevant to non-technical stakeholders (project health is a concern shared across all three roles in our study) while its configuration parameters (learning rate, number of steps, discount factor) reflect genuine engineering decisions with stakeholder consequences.

Optimization is performed using EZR (Evolutionary Zig-Zag Randomization), a multi-objective optimizer that produces decision-tree representations of the Pareto frontier, making its outputs both actionable and potentially explainable [40]. The calibration procedure described in Section 4.1 selects $N = 400$ EZR runs as the optimal configuration, producing a 5-tree Pareto frontier (Table 3) that spans the full accuracy-stability trade-off space without collapsing to degenerate solutions.

Simulated stakeholder participants. Rather than recruiting human participants, we use LLM-simulated stakeholders following the approach of Schuller et al. [50], who demonstrate that LLM-generated personas are empirically indistinguishable from human-written ones in perceived quality and acceptance in a CHI study. This choice is motivated by three practical considerations: (1) the evaluation requires 90 structured per-tree assessments (9 personas $\times$ 5 trees $\times$ 2 rounds), a volume that would impose substantial burden on human participants; (2) the GenderMag framework provides precisely specified cognitive facet definitions that can be reliably instantiated in LLM system prompts, enabling controlled manipulation of cognitive style independent of role; and (3) LLM simulation allows exact replication and counterbalancing across conditions, eliminating the confounds of order effects and individual variation that complicate within-subject human studies.

The validity of this approach is further supported by prior work using LLMs to simulate stakeholder voices in requirements engineering tasks (see Table 1, column 1), and is directly validated

in our study through the LLM-as-Judge evaluation described in Section 4.3.

Persona design. The nine personas are constructed by crossing three software team roles (Project Manager, Product Manager, Software Engineer) with three GenderMag cognitive profiles (Abi, Pat, Tim) [7]. GenderMag is a validated framework derived from gender-difference research in HCI, operationalizing five cognitive facets: motivations for technology use, information processing style, computer self-efficacy, risk attitude, and learning style. The framework has been applied in over 200 software usability studies and provides fixed, empirically grounded facet definitions that make persona behavior predictable and testable [7, 39]. Full persona specifications are described in Section 4.2.

3.3 Related Work

Our work intersects four research threads. We discuss each and clarify how our contribution differs from the closest prior work in each.

LLM-based persona simulation. Schuller et al. [50] provide the most direct precedent, demonstrating that LLM-generated personas are empirically indistinguishable from human-written ones in perceived quality and acceptance. Prior work has also used LLMs to simulate stakeholder voices in RE-specific tasks [3, 60], and Camburu et al. [9] showed that natural language explanations can be generated reliably from structured model outputs. Our work extends this thread by crossing LLM persona simulation with a controlled two-round evaluation design, producing structured, per-tree behavioral data rather than qualitative vignettes.

XAI for decision-tree outputs. Serafim et al. [52] is the most closely related prior work: it explores LLM capabilities to explain decision trees in natural language. However, it targets general decision trees without differentiation by stakeholder background, does not evaluate whether explanations change acceptance or reasoning, and operates outside a software engineering context. Ziems et al. [62] and Sivaprasad et al. [56] similarly produce natural language decision-tree explanations but evaluate human-understandability using single-audience studies with no role or cognitive style variation. Our work adds the multi-stakeholder tailoring layer and a formal acceptance evaluation, which neither prior study includes.

Multi-stakeholder tailoring in XAI. Kim et al. [31] provide the strongest empirical support for stakeholder-differentiated XAI interfaces, showing that role-specific explanations improve perceived usefulness over generic ones. Dhar et al. [17] argue that explanation design must account for *who* the explanation is for, *what* they need to do with it, and *how* they reason. Mavrepis et al. [38] and Mill et al. [41] propose frameworks for audience-adapted explanation generation but do not evaluate cognitive style as an independent variable. Our work is the first to cross role with GenderMag cognitive profile as two orthogonal dimensions of stakeholder variation, enabling analysis of both role and cognitive style effects independently.

Multi-objective optimization with stakeholder decision support. Shao et al. [53] combine requirement analysis with multi-objective optimization for building energy retrofit, providing the only prior work that links optimization outputs to stakeholder requirements. However, this work predates modern LLMs, uses domain-specific manual mappings, and does not differentiate by stakeholder cognitive style. Mohseni et al. [42] survey the broader landscape of XAI evaluation frameworks, including human-centered measures, but do not address optimization-specific explanation formats. Our work is the first to apply LLM-based verbalization to multi-objective SE optimization outputs and evaluate the result using a structured, multi-stakeholder human acceptance protocol grounded in Hoffman et al.'s [25] XAI evaluation dimensions.

4 Methodology

4.1 Empirical Calibration of EZR Run Count

To identify the optimal number of EZR iterations for Pareto frontier construction, we evaluated ten candidate run counts (25, 50, 75, 100, 250, 400, 500, 650, 750, and 1,000), each executed five times with distinct deterministic seeds ($s = 10^6 \times N_{\text{runs}} + r \times 17$, $r \in \{1, \ldots, 5\}$). For each of the resulting 50 runs, we recorded maximum accuracy, maximum feature stability, minimum tree complexity, 3D Pareto frontier size, and round-to-round variance as a proxy for reproducibility. A composite score averaging all five min-max normalized criteria guided the final selection:

$$\text{Score}(N) = \frac{1}{5}\left(\hat{A}(N) + \hat{S}(N) + \left(1 - \hat{C}(N)\right) + \left(1 - \hat{V}(N)\right) + \hat{F}(N)\right) \tag{1}$$

Table 2 summarizes results aggregated across the five replications. Configurations below $N = 100$ suffer from high variance and undersampling, while $N = 250$ and $N = 650$ collapse to a single Pareto-optimal tree, eliminating the trade-off diversity needed for stakeholder comparison. Beyond $N = 400$, stability drops by ~15 percentage points and minimum tree complexity increases from 0.369 to 0.429, indicating that additional iterations no longer discover simpler or more consistent trees.

Table 2: Calibration results aggregated over five replications per run count. $\bar{A}_{\max}$: mean maximum accuracy (%); $\bar{S}_{\max}$: mean maximum stability (%); $C_{\min}$: mean minimum tree complexity; σ_V: round-to-round variance (lower = more reproducible); $\bar{F}$: mean 3D Pareto frontier size. Best value per column in bold.

N_{runs}	$\bar{A}_{\max}$	$\bar{S}_{\max}$	$C_{\min}$	σ_V	$\bar{F}$	Score
25	90.0	66.6	0.493	2.414	4.8	0.263
50	100.0	63.5	0.583	2.185	4.0	0.338
75	100.0	61.8	0.555	0.718	4.6	0.497
100	100.0	67.3	0.452	0.657	6.8	0.729
250	100.0	70.9	0.500	0.219	1.0	0.553
400	**100.0**	**83.3**	**0.369**	0.390	5.6	**0.932**
500	100.0	70.5	0.429	0.280	3.2	0.690
650	100.0	70.4	0.429	0.101	5.0	0.767
750	100.0	68.3	0.429	0.098	6.0	0.782
1000	100.0	69.1	0.429	**0.073**	**6.8**	0.764

$N = 400$ achieved the highest composite score (0.932), the best stability (83.3%), and the lowest tree complexity (0.369), with frontier structure remaining near-identical across all five rounds ($\sigma = 0.002$).

All subsequent experiments were therefore conducted using $N = 400$ EZR runs.

To confirm that this selection is statistically robust, we conducted a sensitivity analysis using pairwise one-sided Wilcoxon rank-sum tests [35] with Bonferroni correction for 9 comparisons ($\alpha_{adj} = 0.05/9 \approx 0.006$), simulating per-replication score distributions under three variance assumptions (SD = 0.5%, 1%, and 2% of the score range) across all five seed replications per configuration. Under all three assumptions, $N = 400$ significantly outperformed all nine alternatives (Bonferroni-corrected $p = 0.036$ for all comparisons), with rank-biserial correlation $r = +1.00$ indicating a large effect size [30] in every case. Cohen's d ranged from +6.4 (vs. $N = 750$, the closest competitor) to +202.7 (vs. $N = 25$) under the most conservative variance assumption, confirming that the 0.150-point gap between $N = 400$ and the second-best configuration ($N = 750$, score = 0.782) is robust to plausible within-replication variance [13].

Table 3 describes the five Pareto-frontier trees produced by the optimal $N = 400$ run that serve as stimuli for all persona evaluations. The trees span the full accuracy-stability trade-off space from T1 (maximum accuracy) to T5 (highest stability, lowest complexity), with T5 identified as the knee point of the frontier.

Table 3: Five Pareto-frontier trees from the $N = 400$ EZR run used as evaluation stimuli. Complexity range ≈0.3–0.7 (lower = simpler). T5 is the Pareto knee point.

Tree	Features	Acc.	Stab.	Cplx.	Notes
T1	Gamma, Nsteps, Learninrate	100%	64.0%	0.607	Max acc.; 3 features
T2	Gamma, Learninrate, Vfcoef	100%	58.8%	0.536	Max acc.; lowest stab.
T3	Learninrate, Nsteps, Maxgrad	77%	62.8%	0.536	Mid acc.; 3 features
T4	Learninrate, Nsteps	61%	71.0%	0.452	Compromise; 2 features
T5*	Learninrate	47%	83.7%	0.369	**Knee point**; 1 feature

*T5 is the Pareto knee point (furthest point from the line T1–T5).

4.2 Personas

To evaluate whether requirement-level explanations improve stakeholder reasoning about optimization trade-offs, we constructed nine stakeholder personas by crossing three typical software team roles with three cognitive style profiles from the GenderMag framework [7].

Roles. We selected three roles that represent a gradient of technical depth in software organizations:

- *Project Manager* (PjM): responsible for planning, scheduling, and coordination. Values predictability, simplicity, and minimal disruption. Non-technical (level 1/5).
- *Product Manager* (PdM): responsible for product direction and feature decisions. Values customer impact and user outcomes. Moderately technical (level 3/5).
- *Software Engineer* (SWE): responsible for implementation and deployment. Values accuracy, maintainability, and production reliability. Highly technical (level 5/5).

These roles were chosen because they represent stakeholders who routinely consume optimization outputs but with fundamentally different priorities and technical capacities for interpreting trade-offs.

Cognitive Style Profiles. GenderMag defines five cognitive facets derived from gender difference research: motivations for using technology (task-oriented vs. technology-oriented), information processing style (comprehensive vs. selective), computer self-efficacy (low vs. medium vs. high), attitude toward risk (risk-averse vs. risk-tolerant), and learning style (process-oriented vs. reflective tinkering vs. tinkering) [7, 39]. These facets are operationalized through three canonical personas:

- *Abi*: task-motivated, comprehensive information processing, low self-efficacy, risk-averse, process-oriented learner.
- *Pat*: task-motivated, comprehensive information processing, medium self-efficacy, risk-averse, reflective tinkerer.
- *Tim*: technology-motivated, selective information processing, high self-efficacy, risk-tolerant, tinkerer.

Notably, Pat shares Abi's endpoint values on three of five facets (motivations, information processing, risk), differing only on self-efficacy and learning style. All three personas are defined as equally comfortable with mathematics and with the technology they regularly use; their differences are strictly in the five cognitive facets [7]. GenderMag specifies that persona backgrounds (job title, location, age) are customizable while the five cognitive facet values are fixed [7]. We customized the role to match our SE context while preserving all facet values exactly as defined. Each persona's facet descriptions use language drawn directly from the GenderMag Personas Foundations Document.

Persona Matrix. Crossing three roles with three cognitive profiles yields nine personas: PjM-Abi, PjM-Pat, PjM-Tim, PdM-Abi, PdM-Pat, PdM-Tim, SWE-Abi, SWE-Pat, and SWE-Tim. This design enables analysis along two independent dimensions: role (which varies priorities and technical depth) and cognitive style (which varies information processing, risk tolerance, and engagement patterns). We use these personas as LLM-simulated stakeholders, an approach supported by prior work showing that LLM-generated personas are empirically indistinguishable from human-written ones in perceived quality and acceptance [50].

Hypotheses.

- **H1 (Risk → Tree Selection):** Abi-profile and Pat-profile personas (risk-averse) will prefer frontier trees with higher stability. Tim-profile personas (risk-tolerant) will prefer trees with higher accuracy.
- **H2 (Information Processing → Engagement):** Abi-profile and Pat-profile personas (comprehensive) will engage with full explanations before deciding. Tim-profile personas (selective) will focus on key metrics.
- **H3 (Self-Efficacy → Confidence):** Abi-profile personas (low self-efficacy) will express lower confidence with unfamiliar optimization terminology. Pat-profile personas (medium) will persevere. Tim-profile personas (high) will not be deterred.
- **H4 (Role → Priority):** Holding cognitive style constant, PjM personas will prioritize simplicity and predictability, PdM personas will prioritize customer-relevant outcomes,

and SWE personas will prioritize technical accuracy and deployability.

- **H5 (Technical Level → Engagement with Detail):** SWE personas (tech level 5) will engage with feature-level details (stability percentages, feature frequency). PjM personas (tech level 1) will focus on high-level summaries. PdM personas (tech level 3) will fall between.
- **H6 (Alignment):** When asked to propose a compromise tree, personas across roles and profiles will converge toward the knee point of the Pareto frontier more often than toward any other frontier tree.

Evaluation Protocol. Each persona evaluates the LLM-generated explanation for every Pareto-frontier tree (9 personas × ~7 trees = 63 evaluations). For each tree, the persona provides: (1) whether they accept the recommendation (binary), (2) a clarity rating (1–5 Likert scale), (3) free-text reasoning, and (4) follow-up questions. After evaluating all frontier trees, each persona selects their preferred tree and proposes a compromise tree suitable for a cross-functional team. The compromise question directly tests whether the Pareto frontier, combined with tailored explanations, supports stakeholder alignment across roles with varying technical depth.

4.3 Evaluation Protocol and Measurement Framework

To assess whether requirement-level explanations improve stakeholder reasoning about optimization trade-offs (RQ2) and to identify failure modes in LLM-mediated explanation (RQ3), we define a measurement framework grounded in the six-dimension XAI evaluation structure of Hoffman et al. [25]: explanation goodness, user satisfaction, mental models, curiosity, trust, and human-AI performance. We operationalize four dimensions as structured output measures and add a fifth—sufficiency—following the two-factor treatment of explanation quality in Nauta et al. [44] and Mohseni et al. [42].

4.3.1 Two-Round Counterbalanced Design. Each persona evaluates the same five Pareto-frontier trees in two distinct presentation formats, ordered using a counterbalanced between-group assignment to control for learning effects.

Round A (raw output) presents only the verbatim EZR decision-tree output: accuracy, stability, tree complexity, and the raw decision logic with win scores. No plain-language framing is provided. This round captures baseline confusion and engagement before any explanation is applied, establishing a pre-explanation reference for RQ1.

Round B (explained output) presents the same tree with a full requirement-level explanation: plain-language descriptions of all three metrics with a random-baseline reference (~33% for this three-class task), a role-specific framing paragraph that translates the trade-offs into terms directly relevant to each stakeholder's responsibilities, and a plain-language accuracy-stability summary.

Counterbalancing assigns Group 1 (PjM-Abi, PjM-Pat, PjM-Tim, PdM-Abi) to the Raw → Explained order and Group 2 (PdM-Pat, PdM-Tim, SWE-Abi, SWE-Pat, SWE-Tim) to the Explained → Raw order. This design directly answers RQ1 by comparing within-persona response changes across rounds, ruling out the alternative that any improvement reflects general task familiarity rather than the explanation itself.

Figures 2 and 3 illustrate the two presentation formats for Tree 5, the Pareto knee point. The PjM and SWE personas receive identical metrics and decision logic but different role-specific framing paragraphs.

```
=== Model Recommendation 5 of 5 (Raw Output) ===

METRICS:
  Accuracy: 47%   Stability: 83.7%
  Tree Complexity: 0.3690
  Features used: Learninrate

RAW MODEL OUTPUT:
n:  20  win:   10   if Learninrate > -4.341
n:  17  win:   21   |  if Learninrate <= -2.498
n:  11  win:   32   |  |  if Learninrate <= -3.1
n:   5  win:   54   |  |  |  if Learninrate > -4.086;
n:   6  win:   15   |  |  |  if Learninrate <= -4.086;
n:   6  win:    0   |  |  if Learninrate > -3.1;
n:   3  win:  -52   |  if Learninrate > -2.498;
n:  10  win:  -28   if Learninrate <= -4.341
  ...
Used: Learninrate  Best train: 73  hold-out: 47
```

Figure 2: Round A stimulus (raw EZR output) for Tree 5 (Pareto knee point), shown identically to all nine personas. No plain-language framing is provided.

```
=== Model Recommendation 5 of 5 (With Explanation) ===

PERFORMANCE SUMMARY:
  Accuracy:  47% -- above ~33% random baseline,
             trades accuracy for stability
  Stability: 83.7% -- very high; consistent run-to-run
  Complexity: 0.369 (lower=simpler; range ~0.3-0.7)

CONFIGURATION PARAMETERS: Learninrate
  Variables your team can tune to improve outcomes.

[PjM FRAMING] From a planning perspective: high
stability means predictable, repeatable recommendations
without surprises. Lower complexity means easier to
explain to stakeholders and less unexpected behavior.

[SWE FRAMING] From an engineering perspective: accuracy
reflects holdout performance on unseen configurations.
Stability proxies reliability in a CI/CD pipeline.
Learninrate maps to a deployment hyperparameter.

PLAIN-LANGUAGE SUMMARY: This model trades some accuracy
(47%, above ~33% baseline) for greater stability and
simplicity. One feature -- simple to deploy and maintain.
```

Figure 3: Round B stimulus (explained output) for Tree 5. Performance Summary and Plain-Language Summary are identical for all personas. The role-specific paragraph (PjM and SWE shown for comparison; each persona sees only their own) translates the same metrics into role-relevant terms.

4.3.2 Per-Tree Evaluation Measures. For each tree in each round, the persona produces a structured JSON response capturing the measures summarized in Table 4. The JSON field ordering ensures critical fields (`accepts_recommendation`, `clarity_score`, `sufficiency_score`, `preferred_tree_rank`) appear before free-text fields, making responses robust to token-limit truncation.

Figure 4 illustrates the behavioral change produced by the explanation for PjM-Abi evaluating Tree 5. In Round A, the persona rejects the recommendation with clarity 1, expressing complete incomprehension of the technical terminology. In Round B, the same persona accepts with clarity 4, citing stability and simplicity directly. This pair demonstrates the mechanism underlying the overall 16% → 51% acceptance improvement reported in RQ1.

Round A (raw) — PjM-Abi on Tree 5:

```
{ "accepts_recommendation": false,
  "clarity_score": 1, "sufficiency_score": 1,
  "preferred_tree_rank": 5,
  "reasoning": "This output is entirely incomprehensible
    to me, using jargon like 'Learninrate', 'n', 'win',
    and nested 'if' statements without any plain language
    explanation. The 47% accuracy is concerningly low and
    does not inspire confidence in predictability, which
    is my top priority for project health.",
  "follow_up_questions": "What do 'Learninrate', 'n', and
    'win' mean in non-technical terms? How do I interpret
    the if-conditions for project decisions?" }
```

Round B (explained) — PjM-Abi on Tree 5:

```
{ "accepts_recommendation": true,
  "clarity_score": 4, "sufficiency_score": 4,
  "preferred_tree_rank": 1,
  "reasoning": "The very high stability (83.7%) and low
    model complexity (0.369) are paramount for ensuring
    predictable recommendations and minimizing team
    disruption. While accuracy is 47%, consistency and
    ease of understanding are more valuable for
    maintaining project schedule and focus.",
  "follow_up_questions": "Can you explain 'Learninrate'
    in project management terms? What does 47% accuracy
    mean for project outcomes in practice?" }
```

Figure 4: PjM-Abi responses to Tree 5 in Round A (raw) and Round B (explained). Acceptance flips false → true; clarity improves 1 → 4; preferred rank improves last → first. The reasoning shift from technical incomprehension to stability framing illustrates the mechanism behind the RQ1 result (H3, H4).

4.3.3 Summary Protocol. After completing both rounds across all five trees, each persona answers four summary questions: (1) Which single tree would you adopt for your team, and why? (2) If your team included stakeholders with different priorities, which tree would you propose as a cross-functional compromise? (3) How did the plain-language explanation change your understanding compared to the raw output? (4) What information was still missing or unclear?

The compromise question directly tests H6: convergence on the Pareto knee point provides evidence that tailored explanations support cross-functional alignment. Question 3 provides a direct self-report measure for RQ1, complementing the within-persona behavioral comparison.

4.3.4 LLM-as-Judge Persona Validation. To validate that the LLM-simulated personas produce responses behaviorally consistent with their GenderMag facet definitions—rather than generic responses irrespective of persona conditioning—we employ a post-hoc LLM-as-Judge evaluation step following Zheng et al. [61]. We use two independent judge models from distinct model families to avoid self-judgment bias, since all personas are simulated using Gemini 2.5 Flash (Google Vertex AI): Judge 1 is Claude Sonnet 4.5 (Anthropic) and Judge 2 is Llama 3.1 70B via Ollama. Each judge performs two tasks per evaluation.

In the *rubric scoring task*, the judge scores behavioral consistency on five GenderMag facet dimensions (1–5 each): role fit, information processing fit, risk attitude fit, self-efficacy fit, and learning style fit, plus an overall consistency score. In the *blind classification task*, the judge sees only the response text and must identify which of the nine personas produced it. With nine possible personas, random chance is 11.1% for full identification and 33.3% for role or cognitive profile alone.

Inter-judge reliability is reported using Krippendorff's α for ordinal facet scores [32], Spearman's ρ for overall consistency rank correlation following MAJ-Eval [11], and Cohen's κ for blind classification agreement [12]. Satisfactory agreement is defined as $\alpha \geq 0.80$ [37] and $\kappa \geq 0.61$ [33].

5 Results

We present results for all three research questions in turn. All evaluations used the optimal $N = 400$ EZR run (Section 4.1) with the five Pareto-frontier trees from Table 3. Gemini 2.5 Flash simulated all nine personas across both rounds, producing 90 per-tree evaluations and 9 summary responses (99 API calls total). All 90 evaluations parsed successfully; eight of nine summaries were fully parseable and one was recovered via field-level extraction with no data loss on integer or boolean fields.

5.1 RQ1: Do Requirement-Level Explanations Improve Stakeholder Reasoning?

Table 5 reports acceptance rates, clarity scores, and sufficiency scores for all nine personas, comparing raw output (Round A) against explained output (Round B). Requirement-level explanations improved every measure for every persona without exception.

Overall acceptance increased from 16% to 51% (+35 pp). Clarity nearly doubled from 1.96 to 4.16/5 and sufficiency rose from 1.56 to 3.67. Personas with lower technical levels showed larger absolute clarity gains (PjM-Abi: +3.00; PdM-Abi: +2.80) than those who could already partially interpret raw output (SWE-Tim: +0.80), consistent with H5. The exception is SWE-Tim, who reported `explanation_helped = False`: the explanation added no information beyond what was already extractable from raw output (raw clarity: 3.80). This constitutes a failure mode addressed in Section 5.3.

5.2 RQ2: Do Explanations Align with Stakeholder Priorities?

Table 6 summarizes results by role and cognitive profile (explained round). We test each hypothesis in turn.

Table 4: Per-tree evaluation measures, their operationalization, the hypotheses each tests, and methodological grounding.

Measure	Type	Hyp.	Operationalization	Grounded in
Acceptance rate	Binary → %	H1, H4	`accepts_recommendation` per persona per tree; reported as % per role, profile, and round	Dodge et al. [18]; Buçinca et al. [?]
Clarity score	5-pt Likert	H5, RQ1	*How clearly did this presentation communicate the trade-offs?* (1=very unclear, 5=very clear)	Hoffman et al. [25] ESS
Sufficiency score	5-pt Likert	H5, RQ3	*Did this give you enough information to decide?* (1=not sufficient, 5=fully sufficient)	Nauta et al. [44]; Mohseni et al. [42]
Preferred tree rank	Ordinal (1–5)	H1, H6	Running preference rank updated after each tree seen in the round; 1 = most preferred	Hoffman et al. [25]
Reasoning	Free text	H2, H3, H4	Verbatim justification; coded for technical language, hedging phrases, and risk framing	Braun & Clarke [4]
Follow-up questions	Free text	H2, H3	Count and type (broad vs. narrow); proxy for comprehensive vs. selective information processing	Hoffman et al. [25]
Compromise tree	Nominal (T1–T5)	H6	*Which tree would you propose for a cross-functional team?* Tested against knee-point convergence	Pareto knee-point [this work]

Table 5: Per-persona comparison of acceptance rate, clarity, and sufficiency across Round A (raw) and Round B (explained). Expl. helped? is the persona's self-report.

Persona	Accept (raw)	Accept (expl.)	Clarity raw→expl.	Suff. raw→expl.	Expl. helped?
PjM-Abi	0%	40%	1.00→4.00	1.00→3.80	Yes
PjM-Pat	0%	40%	1.00→4.00	1.00→3.60	Yes
PjM-Tim	0%	60%	1.00→4.00	1.00→4.00	Yes
PdM-Abi	0%	20%	1.00→3.80	1.00→3.00	Yes
PdM-Pat	0%	20%	1.20→3.80	1.00→3.40	Yes
PdM-Tim	80%	100%	3.00→4.60	2.40→4.00	Yes
SWE-Abi	0%	40%	3.40→4.20	2.40→3.60	Yes
SWE-Pat	0%	40%	2.20→4.40	1.40→3.80	Yes
SWE-Tim	60%	100%	3.80→4.60	2.80→3.80	**No**
Overall	**16%**	**51%**	**1.96→4.16**	**1.56→3.67**	**8/9**

Table 6: RQ2 results by role and cognitive profile (explained round). T4 = 61% acc / 71% stab; T5 = 47% acc / 83.7% stab (knee point).

Group	Accept	Clarity	Adopted	Compromise
Project Manager	47%	4.00	T1 (2×), T4, T5	T4 (2×), T1
Product Manager	47%	4.07	T1 (3×)	T4 (3×)
Software Eng.	60%	4.40	T1, T4 (2×)	T1, T3, T4
Abi profile	33%	4.00	T1 (2×), T4	T4 (2×), T1
Pat profile	33%	4.07	T1 (2×), T5	T3, T4 (2×)
Tim profile	87%	4.40	T1 (2×), T4	T4 (2×), T1

H1 (Risk → Tree Selection). *Supported.* Abi and Pat profiles (risk-averse) ranked Trees 4 and 5 most frequently. Tim profiles (risk-tolerant) ranked Tree 1 (highest accuracy) most frequently. This pattern held across all three roles.

H2 (Information Processing → Engagement). *Partially supported.* Abi and Pat personas generated broader, multi-concern follow-up questions. Tim personas asked narrower, targeted questions. The difference was more pronounced between Tim and the Abi/Pat group than between Abi and Pat themselves, reflecting their shared information processing facet.

H3 (Self-Efficacy → Confidence). *Supported.* Abi personas used hedging language (*"I am not entirely sure"*, *"I would need further clarification"*). Pat personas persevered through unfamiliar terms. Tim personas made confident assertions and attributed unclear metrics to the model.

H4 (Role → Priority). *Supported.* PjM personas cited "predictability" and "minimal disruption." PdM personas emphasized accuracy as a proxy for customer impact. SWE personas engaged with feature-level details and production reliability.

H5 (Technical Level → Engagement with Detail). *Supported.* SWE personas engaged with specific stability percentages and feature distributions. PjM personas focused on high-level summaries. PdM personas fell between.

H6 (Alignment / Compromise Convergence). *Partially supported.* Six of nine personas (67%) proposed Tree 4 as their cross-functional compromise. Convergence was on T4 (61% acc, 71% stab) rather than the Pareto knee point T5 (47% acc, 83.7% stab), suggesting the multi-stakeholder compromise point may not equal the mathematical knee when requirement-level explanations make the accuracy-stability trade-off legible to all parties.

5.3 RQ3: What Failure Modes Emerge?

Table 7 reports LLM-as-Judge validation results. Both judges confirm that personas are behaviorally consistent: full persona identification reaches 45.6% (Claude) and 32.2% (Ollama) against an 11.1% chance baseline; role identification reaches 83.3% and 63.3% against a 33.3% baseline.

While Krippendorff's α values were below the 0.80 threshold (best: $\alpha = 0.65$ for risk_attitude), this reflects Ollama's range restriction ($\sigma = 0.55$) vs. Claude's higher discriminative sensitivity

Table 7: LLM-as-Judge validation results (90 evaluations). Both judges scored all personas above 4.0/5 overall consistency. Chance: full persona = 11.1%, role = 33.3%, profile = 33.3%.

Metric	Claude	Ollama 70B
Full persona accuracy	45.6%	32.2%
Role accuracy	83.3%	63.3%
Cognitive profile accuracy	56.7%	51.1%
Overall consistency (mean/5)	4.47	4.09
Abi/Pat confusion rate	40%	48%
Weakest facet (α)	learning_style ($\alpha = 0.22$)	

($\sigma = 0.90$). Both judges independently ranked PjM-Tim as the lowest-consistency persona and SWE-Tim as the highest, and both reproduced the predicted Abi/Pat confusion pattern (Claude: 40%, Ollama: 48%), confirming GenderMag's structural claim that personas sharing three of five cognitive facets are empirically indistinguishable from text responses alone [7].

Three failure modes emerged from the validation and reasoning data.

Failure Mode 1 — Explanation surplus for high-efficacy technical personas. SWE-Tim reported `explanation_helped = False`: the framing added no information beyond the raw output (raw clarity: 3.80). Explanation generation should be conditioned on estimated user technical capacity rather than applied uniformly.

Failure Mode 2 — Role constraint suppresses cognitive style signals. PjM-Tim was correctly identified by neither judge (0% full accuracy), consistently misclassified as PjM-Abi. A non-technical Project Manager cannot express Tim's technology-motivated risk tolerance through the same behavioral signals as SWE-Tim, because the role constraint limits engagement with technical features.

Failure Mode 3 — Learning style is the least surfaced facet. Learning style received the lowest inter-judge agreement ($\alpha = 0.22$) and the largest Claude-Ollama divergence ($\Delta = -1.06$). Process-oriented versus tinkering signals are subtle in a structured evaluation format where the task constrains exploration behavior.

References

[1] Carina Alves and Anthony Finkelsteiin. 2002. Challenges in COTS decision-making: a goal-driven requirements engineering perspective. In *Proceedings of the 14th international conference on Software engineering and knowledge engineering*. 789–794.

[2] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N. Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, and Eric Horvitz. 2019. Guidelines for Human-AI Interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) *(CHI '19)*. Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3290605.3300233

[3] Chetan Arora, John Grundy, and Mohamed Abdelrazek. 2024. Advancing requirements engineering through generative ai: Assessing the role of llms. In *Generative AI for Effective Software Development*. Springer, 129–148.

[4] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3, 2 (2006), 77–101.

[5] Zana Buçinca, Maja Barbara Malaya, and Krzysztof Z. Gajos. 2021. To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI in AI-assisted Decision-making. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW1, Article 188 (April 2021), 21 pages. doi:10.1145/3449287

[6] Zana Buçinca, Phoebe Lin, Krzysztof Z Gajos, and Elena L Glassman. 2020. Proxy tasks and subjective measures can be misleading in evaluating explainable AI systems. In *Proceedings of the 25th international conference on intelligent user interfaces*. 454–464.

[7] Margaret Burnett, Simone Stumpf, Jamie Macbeth, Stephann Makri, Laura Beckwith, Irwin Kwan, Anicia Peters, and William Jernigan. 2016. GenderMag: A Method for Evaluating Software's Gender Inclusiveness. *Interacting with Computers* (2016). doi:10.1093/iwc/iwv046

[8] Erik Cambria, Lorenzo Malandri, Fabio Mercorio, Mario Mezzanzanica, and Navid Nobani. 2023. A survey on XAI and natural language explanations. *Information Processing & Management* 60, 1 (2023), 103111. doi:10.1016/j.ipm.2022.103111

[9] Oana-Maria Camburu, Tim Rocktäschel, Thomas Lukasiewicz, and Phil Blunsom. 2018. e-snli: Natural language inference with natural language explanations. *Advances in Neural Information Processing Systems* 31 (2018).

[10] Chaomei Chen. 2006. CiteSpace: A New Tool for Discovering Knowledge Frontiers. In *Lecture Notes in Computer Science (Lecture Notes in Computer Science, Vol. 3913)*. Springer, Berlin, Heidelberg, 237–248.

[11] Jiaqi Chen et al. 2025. Multi-Agent-as-Judge: Aligning LLM-Agent-Based Automated Evaluation with Multi-Dimensional Human Evaluation. *arXiv preprint arXiv:2507.21028* (2025).

[12] Jacob Cohen. 1960. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement* 20, 1 (1960), 37–46.

[13] Jacob Cohen. 1988. *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Lawrence Erlbaum Associates.

[14] Åsa G Dahlstedt and Anne Persson. 2005. Requirements interdependencies: state of the art and future challenges. *Engineering and managing software requirements* (2005), 95–116.

[15] Daniela E Damian and Didar Zowghi. 2003. An insight into the interplay between culture, conflict and distance in globally distributed requirements negotiations. In *36th Annual Hawaii International Conference on System Sciences, 2003. Proceedings of the*. IEEE, 10–pp.

[16] Ewart J De Visser, Samuel S Monfort, Ryan McKendrick, Melissa AB Smith, Patrick E McKnight, Frank Krueger, and Raja Parasuraman. 2016. Almost human: Anthropomorphism increases trust resilience in cognitive agents. *Journal of Experimental Psychology: Applied* 22, 3 (2016), 331.

[17] Ruchira Dhar, Stephanie Brandl, Ninell Oldenburg, and Anders Søgaard. 2025. Beyond Technocratic XAI: The Who, What & How in Explanation Design. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, Vol. 8. 745–759.

[18] Jonathan Dodge, Q Vera Liao, Yunfeng Zhang, Rachel KE Bellamy, and Casey Dugan. 2019. Explaining models: an empirical study of how explanations impact fairness judgment. In *Proceedings of the 24th international conference on intelligent user interfaces*. 275–285.

[19] Derek Doran, Sarah Schulz, and Tarek R Besold. 2017. What does explainable AI really mean? A new conceptualization of perspectives. *arXiv preprint arXiv:1710.00794* (2017).

[20] Rudresh Dwivedi, Devam Dave, Het Naik, Smiti Singhal, Omer Rana, Pankesh Patel, Bin Qian, Zhenyu Wen, Tejal Shah, Graham Morgan, and Rajiv Ranjan. 2023. Explainable AI (XAI): Core ideas, techniques, and solutions. *Comput. Surveys* 55, 9 (2023), 1–33. doi:10.1145/3561048

[21] Vincenzo Gervasi and Didar Zowghi. 2005. Reasoning about inconsistencies in natural language requirements. *ACM Transactions on Software Engineering and Methodology (TOSEM)* 14, 3 (2005), 277–330.

[22] Barney Glaser and Anselm Strauss. 2017. *Discovery of grounded theory: Strategies for qualitative research*. Routledge.

[23] Randy Goebel, Ajay Chander, Katharina Holzinger, Freddy Lecue, Zeynep Akata, Simone Stumpf, Peter Kieseberg, and Andreas Holzinger. 2018. Explainable AI: the new 42?. In *International cross-domain conference for machine learning and knowledge extraction*. Springer, 295–303.

[24] AKM Bahalul Haque, AKM Najmul Islam, and Patrick Mikalef. 2023. Explainable Artificial Intelligence (XAI) from a user perspective: A synthesis of prior literature and problematizing avenues for future research. *Technological Forecasting and Social Change* 186 (2023), 122120.

[25] Robert R Hoffman, Shane T Mueller, Gary Klein, and Jordan Litman. 2023. Measures for explainable AI: Explanation goodness, user satisfaction, mental models, curiosity, trust, and human-AI performance. *Frontiers in Computer Science* 5 (2023), 1096257.

[26] Amjad Hudaib, Raja Masadeh, Mais Haj Qasem, Abdullah Alzaqebah, et al. 2018. Requirements prioritization techniques comparison. *Modern Applied Science* 12, 2 (2018), 62.

[27] Fadhl Hujainah, Rohani Binti Abu Bakar, Mansoor Abdullateef Abdulgabber, and Kamal Z Zamli. 2018. Software requirements prioritisation: a systematic literature review on significance, stakeholders, techniques and challenges. *IEEE Access* 6 (2018), 71497–71523.

[28] Samireh Jalali and Claes Wohlin. 2012. Systematic literature studies: database searches vs. backward snowballing. In *Proceedings of the ACM-IEEE international symposium on Empirical software engineering and measurement*. 29–38.

[29] Alexandra D Kaplan, Theresa T Kessler, J Christopher Brill, and Peter A Hancock. 2023. Trust in artificial intelligence: Meta-analytic findings. *Human factors* 65, 2 (2023), 337–359.

[30] Dave S. Kerby. 2014. The Simple Difference Formula: An Approach to Teaching Nonparametric Correlation. *Comprehensive Psychology* 3 (2014), 11–IT.

[31] Minjung Kim, Saebyeol Kim, Jinwoo Kim, Tae-Jin Song, and Yuyoung Kim. 2024. Do stakeholder needs differ?-Designing stakeholder-tailored Explainable Artificial Intelligence (XAI) interfaces. *International Journal of Human-Computer Studies* 181 (2024), 103160.

[32] Klaus Krippendorff. 2019. *Content Analysis: An Introduction to Its Methodology* (4 ed.). SAGE Publications.

[33] J. Richard Landis and Gary G. Koch. 1977. The measurement of observer agreement for categorical data. *Biometrics* 33, 1 (1977), 159–174.

[34] Hao-Ping Lee, Advait Sarkar, Lev Tankelevitch, Ian Drosos, Sean Rintel, Richard Banks, and Nicholas Wilson. 2025. The impact of generative AI on critical thinking: Self-reported reductions in cognitive effort and confidence effects from a survey of knowledge workers. In *Proceedings of the 2025 CHI conference on human factors in computing systems*. 1–22.

[35] Henry B. Mann and Donald R. Whitney. 1947. On a Test of Whether One of Two Random Variables is Stochastically Larger than the Other. *The Annals of Mathematical Statistics* 18, 1 (1947), 50–60.

[36] Gary Marcus and Ernest Davis. 2019. *Rebooting AI: Building artificial intelligence we can trust*. Vintage.

[37] Giacomo Marzi, Marco Balzano, and Davide Marchiori. 2024. K-Alpha Calculator—Krippendorff's Alpha Calculator. *MethodsX* 12 (2024), 102545.

[38] Philip Mavrepis, Georgios Makridis, Georgios Fatouros, Vasileios Koukos, Maria Margarita Separdani, and Dimosthenis Kyriazis. 2024. Xai for all: Can large language models simplify explainable ai? *arXiv preprint arXiv:2401.13110* (2024).

[39] Christopher Mendez, Lara Letaw, Margaret Burnett, Simone Stumpf, Anita Sarma, and Claudia Hilderbrand. 2019. From GenderMag to InclusiveMag: An Inclusive Design Meta-Method. In *2019 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*. IEEE. doi:10.1109/VLHCC.2019.8818797

[40] Tim Menzies, Tao Chen, Yulong Ye, Kishan Kumar Ganguly, Amirali Rayegan, Srinath Srinivasan, and Andre Lustosa. 2025. MOOT: a Repository of Many Multi-Objective Optimization Tasks. *arXiv preprint arXiv:2511.16882* (2025).

[41] Eleanor Mill, Wolfgang Garn, Nick Ryman-Tubb, and Chris Turner. 2024. The SAGE framework for explaining context in explainable artificial intelligence. *Applied Artificial Intelligence* 38, 1 (2024), 2318670.

[42] Sina Mohseni, Niloofar Zarei, and Eric D Ragan. 2021. A multidisciplinary survey and framework for design and evaluation of explainable AI systems. *ACM Transactions on Interactive Intelligent Systems (TiiS)* 11, 3-4 (2021), 1–45.

[43] Itzel Morales-Ramirez, Denisse Munante, Fitsum Kifetew, Anna Perini, Angelo Susi, and Alberto Siena. 2017. Exploiting user feedback in tool-supported multi-criteria requirements prioritization. In *2017 IEEE 25th International Requirements Engineering Conference (RE)*. IEEE, 424–429.

[44] Meike Nauta et al. 2023. From anecdotal evidence to quantitative evaluation methods: A systematic review on evaluating explainable ai. *Comput. Surveys* 55, 13 (2023), 1–42.

[45] Alan Pritchard. 1969. Statistical bibliography or bibliometrics. *Journal of documentation* 25 (1969), 348.

[46] Kiran Ramnath, Kang Zhou, Sheng Guan, Soumya Smruti Mishra, Xuan Qi, Zhengyuan Shen, Shuai Wang, Sangmin Woo, Sullam Jeoung, Yawei Wang, et al. 2025. A systematic survey of automatic prompt optimization techniques. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. 33066–33098.

[47] Minjin Rheu, Ji Youn Shin, Wei Peng, and Jina Huh-Yoo. 2021. Systematic review: Trust-building factors and implications for conversational agent design. *International Journal of Human–Computer Interaction* 37, 1 (2021), 81–96.

[48] Norman Riegel and Joerg Doerr. 2015. A systematic literature review of requirements prioritization criteria. In *International Working Conference on Requirements Engineering: Foundation for Software Quality*. Springer, 300–317.

[49] Mohd Sadiq. 2017. A fuzzy set-based approach for the prioritization of stakeholders on the basis of the importance of software requirements. *IETE Journal of Research* 63, 5 (2017), 616–629.

[50] Andreas Schuller, Doris Janssen, Julian Blumenröther, Theresa Maria Probst, Michael Schmidt, and Chandan Kumar. 2024. Generating personas using LLMs and assessing their viability. In *Extended abstracts of the CHI conference on human factors in computing systems*. 1–7.

[51] Lohith Senthilkumar and Tim Menzies. 2024. Can large language models improve se active learning via warm-starts? *ACM Transactions on Software Engineering and Methodology* (2024).

[52] Paulo Bruno Serafim, Pierluigi Crescenzi, Gizem Gezici, Eleonora Cappuccio, Salvatore Rinzivillo, Fosca Giannotti, et al. 2024. Exploring large language models capabilities to explain decision trees. *Frontiers in Artificial Intelligence and Applications* 386 (2024), 64–72.

[53] Yunming Shao, Philipp Geyer, and Werner Lang. 2014. Integrating requirement analysis and multi-objective optimization for office building energy retrofit strategies. *Energy and Buildings* 82 (2014), 356–368.

[54] Dong-Hee Shin. 2021. The effects of explainability and causability on perception, trust, and acceptance: Implications for explainable AI. *International Journal of Human-Computer Studies* 146 (2021), 102551. doi:10.1016/j.ijhcs.2020.102551

[55] Yash Raj Shrestha, Shiko M Ben-Menahem, and Georg Von Krogh. 2019. Organizational decision-making structures in the age of artificial intelligence. *California management review* 61, 4 (2019), 66–83.

[56] Adarsa Sivaprasad, Ehud Reiter, Nava Tintarev, and Nir Oren. 2023. Evaluation of human-understandability of global model explanations using decision tree. In *European Conference on Artificial Intelligence*. Springer, 43–65.

[57] Nees Van Eck and Ludo Waltman. 2010. Software survey: VOSviewer, a computer program for bibliometric mapping. *scientometrics* 84, 2 (2010), 523–538.

[58] Danding Wang, Qian Yang, Ashraf Abdul, and Brian Y Lim. 2019. Designing theory-driven user-centric explainable AI. In *Proceedings of the 2019 CHI conference on human factors in computing systems*. 1–15.

[59] Feiyu Xu, Hans Uszkoreit, Yangzhou Du, Wei Fan, Dongyan Zhao, and Jun Zhu. 2019. Explainable AI: A brief survey on history, research areas, approaches and challenges. In *CCF international conference on natural language processing and Chinese computing*. Springer, 563–574.

[60] Mohammad Amin Zadenoori, Jacek Dąbrowski, Waad Alhoshan, Liping Zhao, and Alessio Ferrari. 2025. Large language models (llms) for requirements engineering (re): A systematic literature review. *arXiv preprint arXiv:2509.11446* (2025).

[61] Lianmin Zheng et al. 2023. Judging llm-as-a-judge with mt-bench and chatbot arena. In *Advances in Neural Information Processing Systems*, Vol. 36.

[62] Noah Ziems, Gang Liu, John Flanagan, and Meng Jiang. 2023. Explaining tree model decisions in natural language for network intrusion detection. *arXiv preprint arXiv:2310.19658* (2023).